

Michael Osiako Afote

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Artificial Intelligence Researcher with a B.Sc. in Computer Science (CGPA 4.43/5.0, top 3% of cohort) and a peer-reviewed publication on ensemble-based fall detection. Delivered six end-to-end AI applications in production spanning healthcare safety, NLP, fairness-aware hiring, and generative image synthesis. Operates across the full Machine Learning lifecycle data exploration, model development, Flask API engineering, and cloud deployment. Experienced in applied research, developer mentorship, and building from scratch in PyTorch and scikit-learn.

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, LangChain, llama-index), R, SQL, Bash, Java, C++, JavaScript, PHP

Frameworks & Tools: Flask, Hugging Face Transformers, MongoDB, Node.js, Excel, Git/GitHub

Competencies: Data Analysis, Statistical Modeling, NLP, Deep Learning, Feature Engineering, MLOps, Cloud Deployment

WORK EXPERIENCE

Freelance AI/ML Researcher

2023 – Present

Independent · Remote

- Conduct literature reviews and exploratory data analysis for client research projects spanning healthcare, NLP, and applied machine learning.
- Fine-tuned GPT-2 and Mistral language models; evaluated generation quality and applied bias mitigation strategies to improve output fairness.
- Prepared and presented technical research reports, translating findings into clear, actionable recommendations for non-technical stakeholders.

Machine Learning Developer & Data Scientist

2022 –2025

GITECH · Kogi State, Nigeria

- Designed, trained, and deployed supervised and unsupervised Machine Learning models across multiple client engagements; maintained all code and documentation on GitHub for reproducibility and team handover.
- Built a content-based meal recommender using TF-IDF and cosine similarity achieving sub-second latency, deployed live at [predictive-meal.onrender.com](#).
- Ran end-to-end exploratory data analysis and feature selection pipelines that identified key behavioural drivers, enabling data-informed product decisions.
- Standardised the team model serialisation workflow (pickle + joblib + scaler versioning), reducing model swap-in time by approximately 60% across projects.

Web Developer & Coding Tutor

2022 –2023

GITECH · Kogi State, Nigeria

- Built and maintained responsive client web applications using HTML, CSS, JavaScript, PHP, Node.js, and ASP.NET.
- Tutored students in front-end development from zero to functional interactive UIs; consistently received positive feedback for teaching clarity.
- Contributed to curriculum design and mentored junior developers on debugging methodology and code review.

Freelance Coding Mentor

2023 – Present

Independent · Remote

- Mentored developers at varying skill levels in Python, ML project workflows, and software engineering best practices.
- Diagnosed and resolved code-level bugs across capstone projects, consistently reducing time-to-completion for mentees.

FEATURED PROJECTS

NaijaDetect — Nigerian Language Hate & Offensive Speech Detector

2024 – 2026

- NLP system detecting hate and offensive speech across Yoruba, Hausa, and Igbo, achieving 90 % average F1-score across all three languages.
- Built with Python and Flask; deployed with real-time AJAX inference rendering results in under 500 ms. Live at hate-speech-detection-wazobia.onrender.com.

Accidental Fall Prediction & Detection System

2023 – 2025

- Designed and implemented an ensemble ML model to predict and detect accidental falls in elderly persons in real time, achieving 94% classification accuracy.
- Integrated automated caretaker notifications on detection, reducing average caretaker response time by approximately 65%.
- Research findings published in the International Journal of Computer Applications (IJCA), April 2025.

Flavr — AI-Powered Meal Recommendation Engine

2024 – 2025

- Developed an ML-based personalised meal recommender providing full nutritional breakdowns and step-by-step cooking instructions.
- Improved recommendation relevance by approximately 40% over keyword-based search; deployed at predictive-meal.onrender.com.

Wofash GAN — Generative Image Synthesis for African Women Fashion

2022 – 2023

- Implemented a Deep Convolutional GAN (DCGAN) to generate realistic African women fashion designs, demonstrating practical application of generative AI in cultural and creative contexts.

EDUCATION

B.Sc. Computer Science

2018 – 2023

Federal University Lokoja, Kogi State, Nigeria

CGPA: 4.43 / 5.0 — Top 3% of Cohort

PUBLICATIONS

Afote, M. O., & Ochepe, F. O. (2025). Accidental Fall Prediction and Detection in Elderly Persons using Ensemble Techniques. *International Journal of Computer Applications*, 186(78), 32–36. DOI: [10.5120/ijca2025924684](https://doi.org/10.5120/ijca2025924684)

PROFESSIONAL AFFILIATIONS

- Data Scientists Network / Data Science Nigeria (DSN) · 2021 – Present
- Nigeria Association of Computing Students (NACOS) · 2018 – 2023

AWARDS & HONOURS

- Top 3% of Graduating Class — Federal University Lokoja, 2023
- Peer-reviewed Journal Publication — Research accepted in IJCA (Impact-indexed), April 2025